CS190C Lec6

Training and Autoregressive Decoding

Overview

- Model Training: Macro System Architecture
- Training modules
  - Data Sampling Module
  - AdamW Optimizer
  - Training Auxiliary Modules
  - Checkpoints
- Training and Decoding
  - Start of Training: Input Parameter Decoding
  - Complete script
  - Autoregressive Decoding

PART1 - Model Training: Macro System Architecture

Overview of Model Training Process

- Determine the total number of iterative optimization steps. For each step:
  - Sample input text and standard output.
  - Forward.
  - Calculate loss function value.
  - Calculate gradients and clipping.
  - Schedule the learning rate.
  - Update parameters.
- It is recommended to output real-time status during the process, such as the current loss function value.

Modules to Implement

- Input & Standard output data $\Rightarrow$ `Data Sampling` Module
- Loss calculation $\Rightarrow$ `Cross-Entropy Loss` Module
- Calculate and process Gradients $\Rightarrow$ `Gradient Clipping` Module
- Parameter Optimization $\Rightarrow$ `Learning Rate Scheduler` Module, `AdamW` Optimizer Module
- `Checkpoint` Module, `Pre-decoding` Module

PART2: Data Sampling Module

Memory Issues with Very Large Corpus

- The corpus will initially be encoded to a long string of numbers using BPE.
- To sample a part of it as a "sentence", a naive method is to read the whole long string into memory, and sample hundreds of numbers.
- If the corpus is extremely large, reading all of them will bring disaster to memory.

Memory Issues with Very Large Corpus

- Is there a solution that reads only specific parts needed from disk to memory?
- `numpy.memmap`: A list that interacts directly with disk space for reading and writing, and which is highly similar to a regular list in other aspects.
- We can implement `class Memmap_Manager`, containing `def save_as_memmap` and `def load_by_range`, for disk-oriented read/write operations.
- The core idea is: Store the whole long string on disk, only read a small part of it into memory according to our needs.

Ideas

We can implement this idea roughly...

- Suppose the whole long string contains $500,000,000$ numbers.
- We can cut it into small chunks, each chunk's length= $500,000$.
- Store these 10000 chunks on disk.
- Suppose we need to read from 2000-th to 8000-th number, what chunks should we load into memory?
  - $490,000$-th to $1,050,000$-th ?
- These two phases correspond to `def save_as_memmap` and `def load_by_range`.

1. `def save_as_memmap`

- Use `BPE_Tokenizer`'s `encode_iterable`, reading only a single number each time.
- Continuously read numbers into a buffer list.
- When the buffer reaches a certain size (e.g., $500,000$ numbers), write it as a whole block to disk.
- Clear the buffer and continue reading and writing blocks until the corpus is finished.
- Count the total number of integer codes in the corpus during the process.

That is: **saving as encoding**.

1. `def save_as_memmap`

```
[CODE EXPUNGED]
```

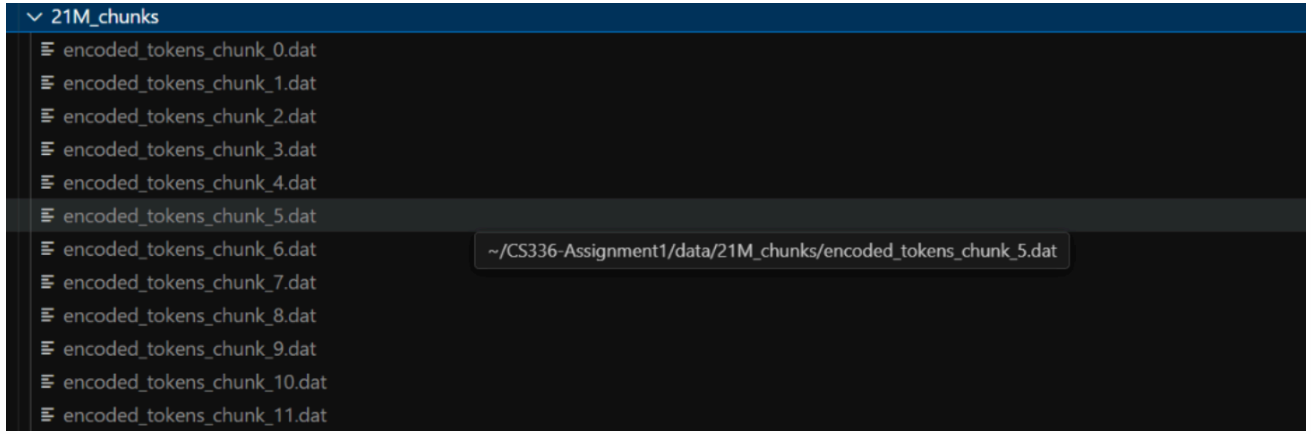
1. `def save_as_memmap`

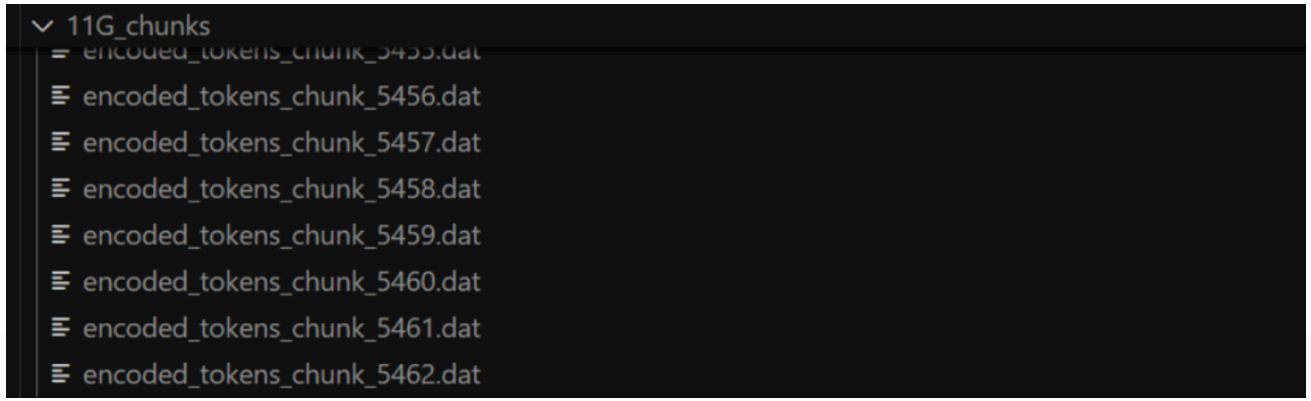
```
[CODE EXPUNGED]
```

For np.memmap:

- Define array type and shape in advance.
- Operate like a regular array, still stored in memory.
- Explicit flush operation writes to disk, freeing the previously allocated memory automatically.

1. `def save_as_memmap`





Attempting to load such a large complete list into memory is catastrophic!

2. `def load_by_range`

Requirement: We need to read all elements in the range `[start_idx, end_idx)` of the encoded list.

- Calculate: Which chunks covered?
- Load only corresponding intervals into memory.
- Operations are consistent with regular lists, except for declaring the `np.memmap` type.

2. `def load_by_range`

`memmap_arr[idx_in_start:idx_in_end]`: Loads this interval from disk into memory.

```
[CODE EXPUNGED]
```

3. Formal Sampling Operation: `class Batch_By_Memmap`

Problem1: How to sample input data?

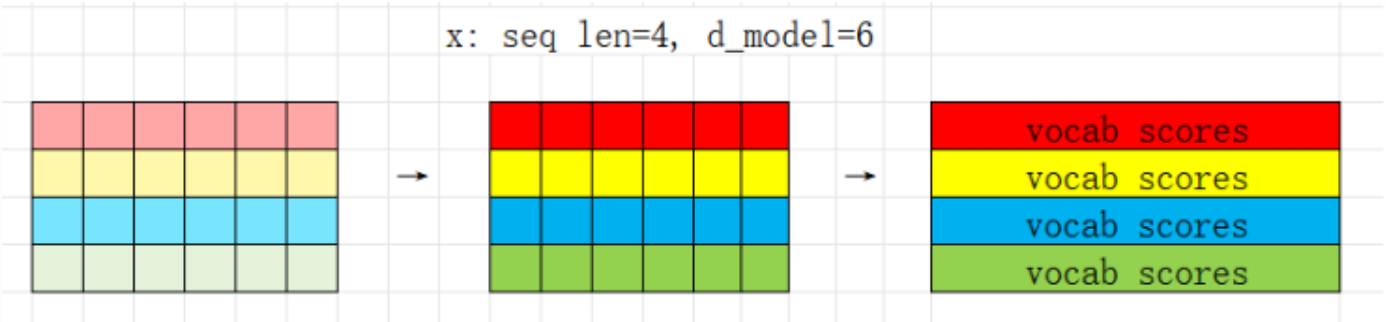
Assuming a corpus length of 10 and a single sampling sequence length of 4:

- Dense sampling: 1234; 2345; 3456; ...
  - Too much data. If the total number of training steps is small (i.e., 3 samples), the second half of the corpus might not be covered.
- Sequential sampling: 1234; 5678; ...
  - Some data is missing. E.g., cannot get 3456 as training input.
- Compromise: Random sampling
  - Randomly pick a start point (ensuring it's <= 7), then continuously sample 4 characters from the start point.
  - Usually sample `batch_size` sequences simultaneously, so randomly pick `batch_size` start points to sample.

3. Formal Sampling Operation: `class Batch_By_Memmap`

Problem2: What is the standard output?

- Recall what the LLM does: each vector is finally linearly projected to `vocab_size`-dimensional scale weights to predict the word at the next position (as the diagram below).
- So the standard output of position-i should be the $(i+1)$-th word in the corpus!
- That is: if the input sequence is 1234, the standard output should be 2345.



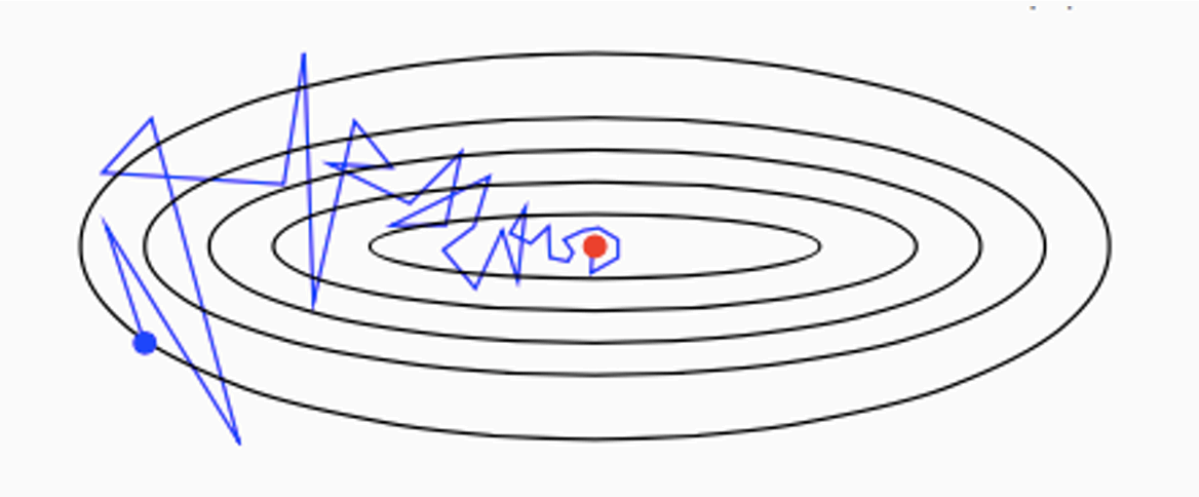
3. Formal Sampling Operation: `class Batch_By_Memmap`

```
[CODE EXPUNGED]
```

PART3: AdamW Optimizer

1. From `SGD` to `AdamW`

SGD: $\theta = \theta - \alpha \nabla L(\theta)$



1. From `SGD` to `AdamW`

What's the defect of SGD?

- Zig-zagging since the gradient may not be stable.
- Shares the same learning rate for all parameter updates.
  - Some parameter, such as word embedding of `the`, may update repeatedly and significantly, leading to possible oscillation.
  - Some parameter, such as word embedding of `pneumoconiosis`, may rarely update, leading to little optimization.

1. From `SGD` to `AdamW`

What `AdamW` does?

- Use first moment `m` (called "**momentum**") to record historical gradient information.
- Use second moment `v` to record historical gradient fluctuation information.
- Use the first and second moments to adjust the parameter update process.

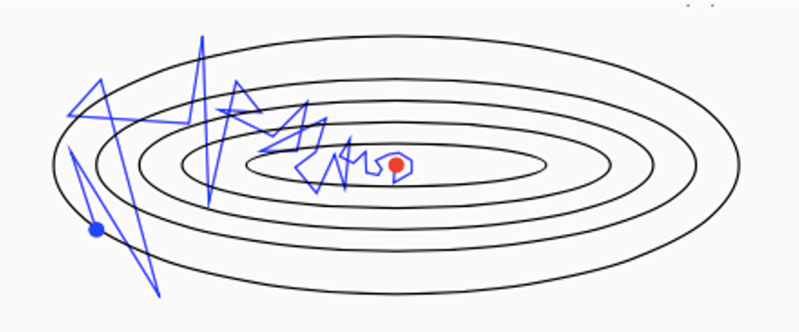
1. From `SGD` to `AdamW`

Let gradient of step t be g_t:

$$m_t = \beta_1(m_{t-1} - g_t) + g_t$$

It means: $m_t = (1-\beta_1)\sum_{\tau=1}^{t}\beta_1^{t-\tau}g_\tau$, which is the exponential weighting of historical gradients.

We use m_t as the "gradient" of step t instead of g_t, reducing zig-zagging of gradient. m_t can now hardly fluctuate as severely.



1. From `SGD` to `AdamW`

$$v_t = \beta_2(v_{t-1} - g_t^2) + g_t^2$$

It means: $v_t = (1-\beta_2)\sum_{\tau=1}^{t}\beta_2^{t-\tau}g_\tau^2$, which is the **exponential weighting** of historical square of gradients.

Additionally, it can be understood as: Variance given $\mathrm{Mean} = 0$ and exponential weights. But what if $\mathrm{Mean} \neq 0$?

Suppose a sequence of gradients with the same mean value: `2,2,2,2` `3,1,3,1`, we find `3,1,3,1` still leads to larger value of v. So it can reflect the fluctuation of historical gradients.

1. From `SGD` to `AdamW`

So how do we use this information to adjust $\theta_t = \theta_{t-1} - \alpha g_t$?

- For the first moment, change g_t to m_t
- How to make use of second moment?
  - If v is large: It means the gradient of the parameter fluctuates significantly, such as for the word embedding of high-frequency words. The update step should be smaller.
  - If v is small: It means the parameter rarely updates, for example, the historical gradients are `0,0,0,1,0,0`. The update step should be larger.

$$\theta_t = \theta_{t-1} - \alpha \frac{m_t}{\sqrt{v_t} + \epsilon}$$

1. From `SGD` to `AdamW`

$$\theta_t = \theta_{t-1} - \alpha \frac{m_t}{\sqrt{v_t} + \epsilon}$$

Is there any hidden problem with it?

When t is small...

- $m_1 = (1-\beta_1)g_1, m_2 = \beta_1(1-\beta_1)g_1 + (1-\beta_1)g_2 \ldots$
- Suppose all $|g_i|$ share the similar scale, so $|m_1| = (1-\beta_1)|g|$, $|m_2| = (1-\beta_1^2)|g|$ $, \ldots, |m_t| = (1-\beta_1^t)|g| \Rightarrow$ The scale is distorted compared with $|g|$
- The same with $|v_t|$: $|v_t| = \sqrt{1-\beta_2^t}|g|$

1. From `SGD` to `AdamW`

Fix the scale distortion: $m_t \Rightarrow \frac{m_t}{1-\beta_1^t}$, $v_t \Rightarrow \frac{v_t}{\sqrt{1-\beta_2^t}}$

That is:

$$\theta_t = \theta_{t-1} - \alpha \frac{\sqrt{1-\beta_2^t}}{1-\beta_1^t} \frac{m_t}{\sqrt{v_t} + \epsilon}$$

We can implement it in engineering:

$$\alpha_t = \alpha \frac{\sqrt{1-\beta_2^t}}{1-\beta_1^t}$$

$$\theta_t = \theta_{t-1} - \alpha_t \frac{m_t}{\sqrt{v_t} + \epsilon}$$

1. From `SGD` to `AdamW`

It is the adjustment of SGD without regularization.

- What if we consider L2-regularization (weight decay)?
- Recall: L2-regularization penalizes large weights to improve generalization.

$$
\begin{aligned}
L_{\text{total}}(\theta) &= L_{\text{task}}(\theta) + \frac{\lambda}{2}\|\theta\|_2^2 \\
\implies \theta &= \theta - \alpha \nabla L_{\text{total}}(\theta) \\
\implies \theta &= \theta - \alpha \nabla L_{\text{task}}(\theta) - \alpha\lambda\theta
\end{aligned}
$$

- However, the regularization complicates the adaptive method in `Adam` when we consider the gradient of L_{total}.
  - `Adam` scales gradients adaptively, L2 regularization is scaled as well, making the effective weight decay parameter-dependent instead of constant λ .

1. From `SGD` to `AdamW`

It is the adjustment of SGD without regularization.

- What if we consider L2-regularization (weight decay)?
- However, the regularization complicates the adaptive method in `Adam`.

So in `AdamW`, we use decoupled weight decay:

$$
\begin{aligned}
\theta_t &= \theta_{t-1} - \alpha \frac{\sqrt{1-\beta_2^t}}{1-\beta_1^t} \frac{m_t}{\sqrt{v_t}+\epsilon} - \alpha\lambda\theta \\
&= \theta_{t-1} - \alpha_t \frac{m_t}{\sqrt{v_t}+\epsilon} - \alpha\lambda\theta
\end{aligned}
$$

1. From `SGD` to `AdamW`

What is the order of magnitude of the update scale?

Assumption: When the training process enters the middle and late stages,

- The gradient $|g|$ can be considered as a sequence with $\mathrm{Mean} = 0$ and $\mathrm{Variation} = \sigma^2 \implies |g| \to \Theta(\sigma)$
- The adaptive learning rate $\alpha_t \to \alpha$

SGD:

$$|\alpha g_t| = \Theta(\alpha\sigma)$$

1. From `SGD` to `AdamW`

Suppose $|g| \to \Theta(\sigma), \alpha_t \to \alpha$, for AdamW:

$$\left|\alpha_t \frac{m_t}{\sqrt{v_t}+\epsilon}\right| = \Theta\left(\alpha_t \frac{\sum_{k=1}^{t} \beta_1^k |g|_{t-k}}{\sqrt{\sum_{k=1}^{t} \beta_2^k |g|_{t-k}^2}}\right)$$

$$|m_t| = \sum_{k=1}^{t} \beta_1^k |g|_{t-k} = \Theta(\sigma)$$

$$|\sqrt{v_t}| = \sqrt{\sum_{k=1}^{t} \beta_2^k |g|_{t-k}^2} = \Theta\left(\sqrt{\sigma^2}\right) = \Theta(\sigma)$$

Then $\left|\alpha_t \frac{m_t}{\sqrt{v_t}+\epsilon}\right| = \Theta(\alpha)$, which is only related to base learning rate.

2. `torch.optim.Optimizer` Base Class

Any implementation of a custom optimizer inherits from the `torch.optim.Optimizer` base class.
The Optimizer base class provides two-level management for all model parameters:

- `self.param_groups`: Group parameters. Each group can set different features like learning rate. (First-level management: parameter grouping, sharing states within the group)
- Each `param_groups` contains at least the default key-value pair: `"params"`, corresponding to a list of parameters. Other key-value pairs like `"lr"` can also be customized.
- For each parameter in the parameter list, various states can also be set, e.g., iteration step `"step"`. (Second-level management: finer-grained states for individual parameters)

2. `torch.optim.Optimizer` Base Class

```
[CODE EXPUNGED]
```

3. `class AdamW_Optimizer`

Algorithm 1 AdamW Optimizer

init(θ) (Initialize learnable parameters)
$m \leftarrow 0$ (Initial value of the first moment vector; same shape as θ)
$v \leftarrow 0$ (Initial value of the second moment vector; same shape as θ)
for $t = 1, \ldots, T$ **do**
  Sample batch of data B_t
  $g \leftarrow \nabla_\theta \ell(\theta; B_t)$ (Compute the gradient of the loss at the current time step)
  $m \leftarrow \beta_1 m + (1 - \beta_1) g$ (Update the first moment estimate)
  $v \leftarrow \beta_2 v + (1 - \beta_2) g^2$ (Update the second moment estimate)
  $\alpha_t \leftarrow \alpha \frac{\sqrt{1-(\beta_2)^t}}{1-(\beta_1)^t}$ (Compute adjusted α for iteration t)
  $\theta \leftarrow \theta - \alpha_t \frac{m}{\sqrt{v+\epsilon}}$ (Update the parameters)
  $\theta \leftarrow \theta - \alpha \lambda \theta$ (Apply weight decay)
end for

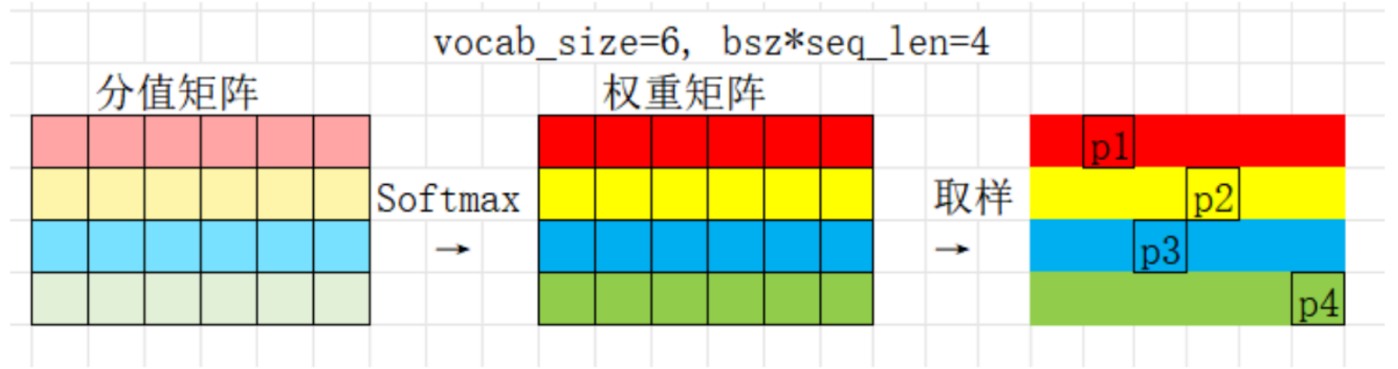
```
[CODE EXPUNGED]
```

PART4: Training Auxiliary Modules

1. `class Cross_Entropy_Calculator`

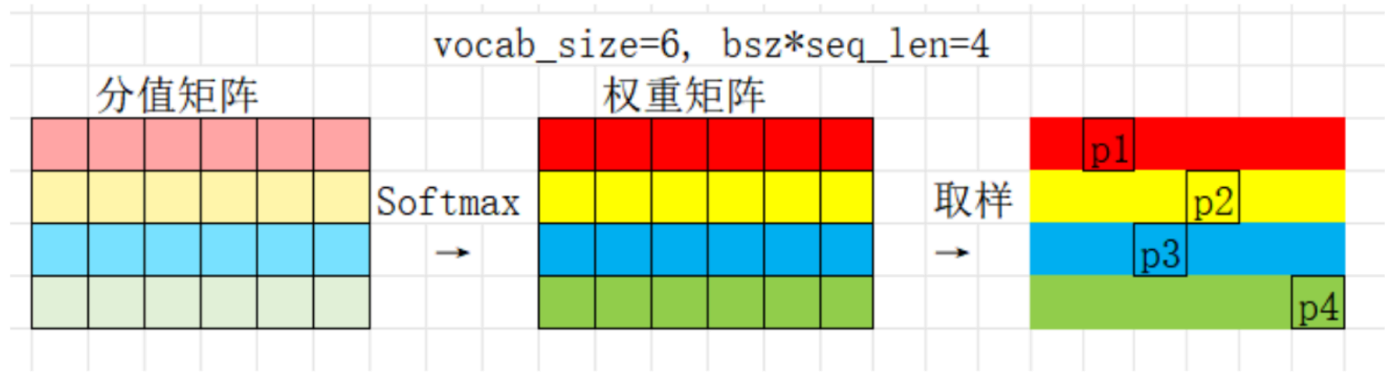
Review: What the LLM does, and what it means.

- Transformer output shape is `[bsz, seq_len, vocab_size]`. There are `(bsz*seq_len)` output predictions, each prediction is "the probability score s_i for each word at the next position".
- Perform Softmax to get the probability distribution p_i of the next word.



1. `class Cross_Entropy_Calculator`

- Take $-\log$ of the probability weight to get $l_i = -\log(p_i)$
- the mean of all l_i is the **cross-entropy loss**.
- $\text{Loss} = \text{Mean}\{-l_i\} = \text{Mean}\{-\log p_i\} = \text{Mean}\{-\log \text{Softmax}(s_i)\}$



1. `class Cross_Entropy_Calculator`

Numerical hazard: What if p_i is too small and approximated as 0?

- Suppose token1's 6-word weight scores are `100, -1, 1, 5, 2, 1000`, and the standard output is word2, with score `-1` and weight `0`.
- Calculation result: $\log 0 = \mathrm{NAN}$

Formula derivation:

$$
\begin{aligned}
\mathrm{logSoftmax}(s_i) &= \log \frac{\exp(s_i)}{\sum \exp(s_k)} \\
&= \log \frac{\exp(s_i - s_{\max})}{\sum \exp(s_k - s_{\max})} \\
&= (s_i - s_{\max}) - \log \left(\sum \exp(s_k - s_{\max}) \right)
\end{aligned}
$$

1. `class Cross_Entropy_Calculator`

```
[CODE EXPUNGED]
```

1. `class Cross_Entropy_Calculator`

```
[CODE EXPUNGED]
```

2. `class Gradient_Clipper`

Suppose the gradient of a parameter is too large?

- SGD: $|\alpha g_t| = \Theta(\alpha|g|)$. It may lead to failure of optimization.
- AdamW: $\left|\alpha_t \frac{m_t}{\sqrt{v_t}+\epsilon}\right| = \Theta(\alpha)$. It seems like there is no problem?

Review: v_t of AdamW: $v_t = (1-\beta_2)\sum_{\tau=1}^{t} \beta_2^{t-\tau} g_\tau^2$

- v_t becomes extremely large over the next hundreds of steps.
- $\alpha_t \frac{m_t}{\sqrt{v_t}+\epsilon} \to 0$, which means the optimization stops abnormally.

So when the gradient is too large, we need to scale it to a small enough level.

2. `class Gradient_Clipper`

Suppose all parameters of a model come from 5 Linear_Transform (5D to 5D), then the model has 5 parameter tensors (all of 5×5 shape).
After one round of backpropagation, suppose the gradients of all 5 parameters are equal to:

```
tensor([[1., 1., 1., 1., 1.],
        [1., 1., 1., 1., 1.],
        [1., 1., 1., 1., 1.],
        [1., 1., 1., 1., 1.],
        [1., 1., 1., 1., 1.]])
```

Let the parameter tensors be p_1, p_2, p_3, p_4, p_5. $\|\nabla p_i\|_2 = \sqrt{25 * 1} = 5$

So the grad norm is $\sqrt{\sum_{i=1}^{5} \|\nabla p_i\|_2^2} = \sqrt{125} = 11.18$

2. `class Gradient_Clipper`

- Assuming the acceptable upper limit is `max_norm` $=0.01$:
- All parameters must be multiplied by the scaling factor $\frac{\text{max_norm}}{g+\epsilon}$ to ensure the total L2 norm of the gradients remains within a reasonable range.
- After clipping, the gradients of each parameter are $11.18/0.01$ times smaller:

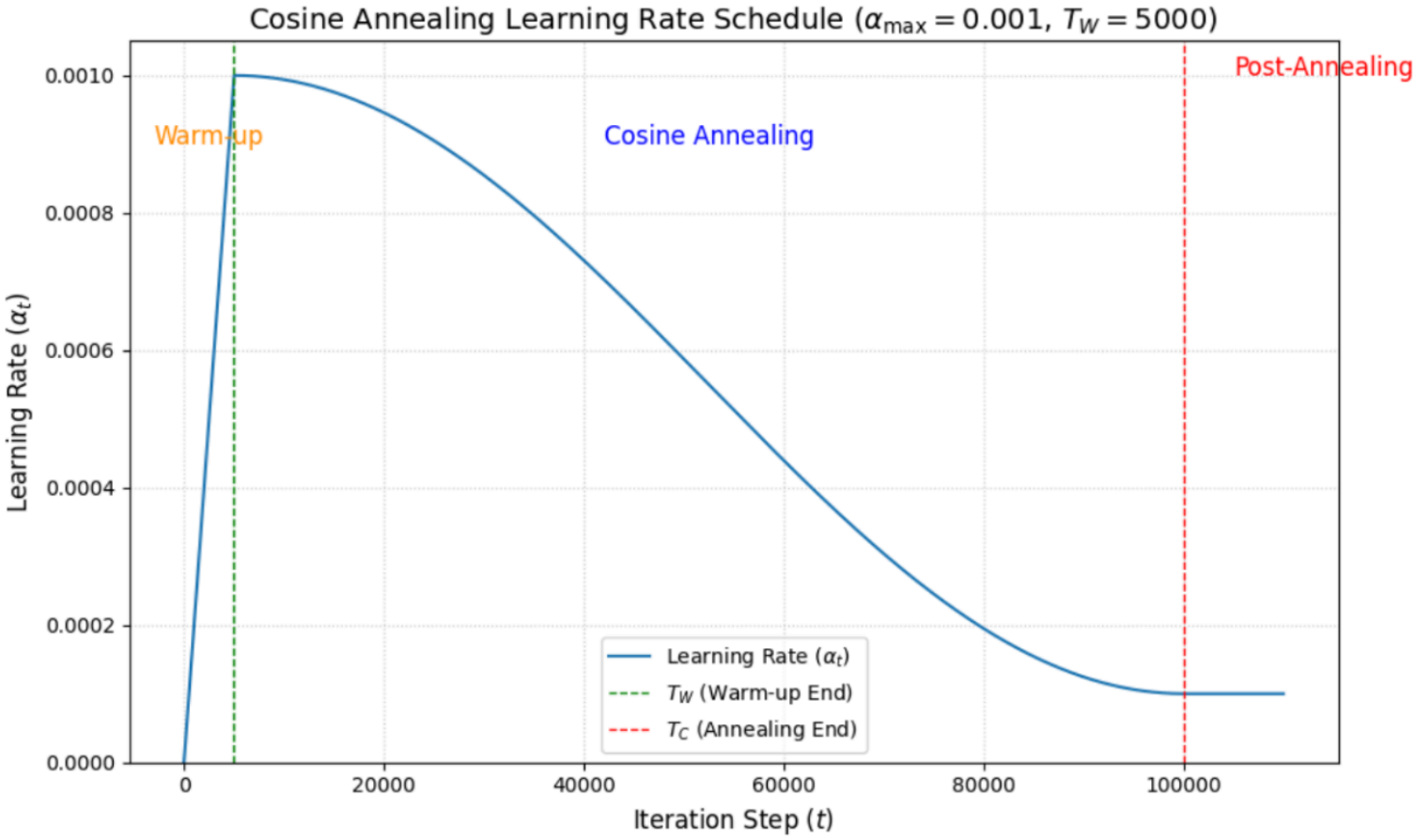
```
tensor([[0.0009, 0.0009, 0.0009, 0.0009, 0.0009],
        [0.0009, 0.0009, 0.0009, 0.0009, 0.0009],
        [0.0009, 0.0009, 0.0009, 0.0009, 0.0009],
        [0.0009, 0.0009, 0.0009, 0.0009, 0.0009],
        [0.0009, 0.0009, 0.0009, 0.0009, 0.0009]])
```

2. `class Gradient_Clipper`

```
[CODE EXPUNGED]
```

3. `class Learning_Rate_Scheduler`

Throughout the model training process, the learning rate is not constant:



3. `class Learning_Rate_Scheduler`

```
[CODE EXPUNGED]
```

PART5: Checkpoints

A hidden problem

Usually we need hours or even days to train a large model...

Can we ensure there's nothing wrong during the whole process?

- For example, the computing power resource you applied for has been preempted, and everything crashed immediately.

To address this issue, we try to save the model every once in a while.

This is `checkpoint`.

What to save and load?

- Model parameters
- Optimizer parameters and states (such as m_t and v_t)
- Current iteration step

We need to get them first, and try to save them as files on disk.

1. How to Get All Parameters?

- PyTorch has a built-in `state_dict()` function that stores all parameters of the PyTorch module in dictionary form.

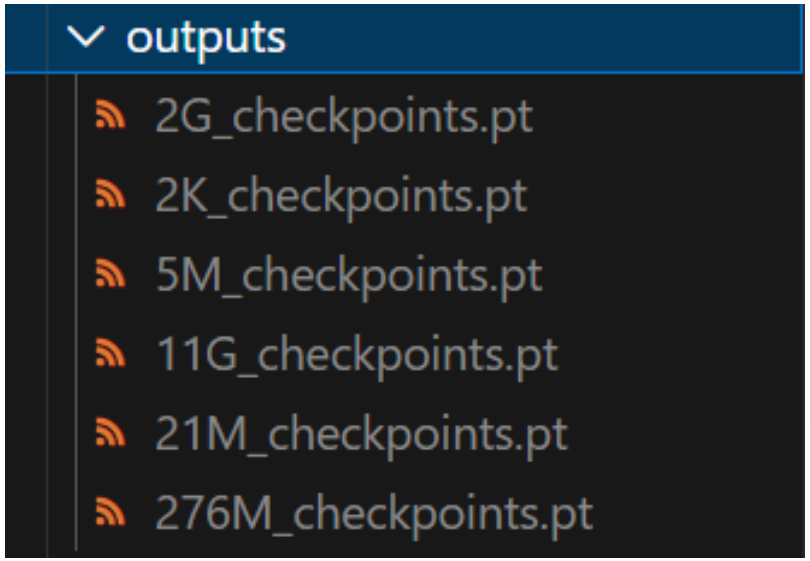
```python
lm = Transformer_LM(
    d_model=512,
    num_heads=8,
    d_ff=2048,
    vocab_size=10000,
    num_layers=2,
    max_seq_length=128,
    theta=10000,
    dtype=torch.float32,
    device="cpu"
)

states = lm.state_dict()
for state_key in states:
    print(state_key, states[state_key].shape)
```

```
transformer_blocks.0.RMSNorm_Attn.g torch.Size([512])
transformer_blocks.0.RMSNorm_FF.g torch.Size([512])
transformer_blocks.0.Multihead_Attn.q_proj.linear_matrix torch.Size([512, 512])
transformer_blocks.0.Multihead_Attn.k_proj.linear_matrix torch.Size([512, 512])
transformer_blocks.0.Multihead_Attn.v_proj.linear_matrix torch.Size([512, 512])
transformer_blocks.0.Multihead_Attn.o_proj.linear_matrix torch.Size([512, 512])
transformer_blocks.0.Multihead_Attn.rope.cos_values torch.Size([1, 128, 32])
transformer_blocks.0.Multihead_Attn.rope.sin_values torch.Size([1, 128, 32])
transformer_blocks.0.Feed_Forward.linear_w1.linear_matrix torch.Size([512, 2048])
transformer_blocks.0.Feed_Forward.linear_w3.linear_matrix torch.Size([512, 2048])
transformer_blocks.0.Feed_Forward.linear_w2.linear_matrix torch.Size([2048, 512])
transformer_blocks.1.RMSNorm_Attn.g torch.Size([512])
transformer_blocks.1.RMSNorm_FF.g torch.Size([512])
transformer_blocks.1.Multihead_Attn.q_proj.linear_matrix torch.Size([512, 512])
transformer_blocks.1.Multihead_Attn.k_proj.linear_matrix torch.Size([512, 512])
transformer_blocks.1.Multihead_Attn.v_proj.linear_matrix torch.Size([512, 512])
transformer_blocks.1.Multihead_Attn.o_proj.linear_matrix torch.Size([512, 512])
transformer_blocks.1.Multihead_Attn.rope.cos_values torch.Size([1, 128, 32])
transformer_blocks.1.Multihead_Attn.rope.sin_values torch.Size([1, 128, 32])
transformer_blocks.1.Feed_Forward.linear_w1.linear_matrix torch.Size([512, 2048])
transformer_blocks.1.Feed_Forward.linear_w3.linear_matrix torch.Size([512, 2048])
transformer_blocks.1.Feed_Forward.linear_w2.linear_matrix torch.Size([2048, 512])
final_norm.g torch.Size([512])
final_layer.linear_matrix torch.Size([512, 10000])
```

2. How to Load All Parameters?

- PyTorch has a built-in `load_state_dict()` function that loads the values of all parameters of the PyTorch module from a state_dict dictionary.
- Prerequisite: Both must match completely!



3. `class Checkpoint_Manager`

- Objects to store: Model parameters, optimizer parameters, current iteration step.

```
[CODE EXPUNGED]
```

3. `class Checkpoint_Manager`

- Objects to load: Model parameters, optimizer parameters, current iteration step.

```
[CODE EXPUNGED]
```

PART6: Start of Training: Input Parameter Decoding

To This Point...

We have sorted out:

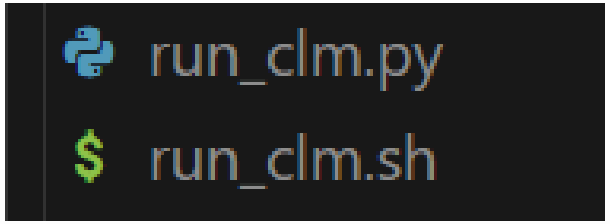
- The architecture of each model module and its required parameters.
- The implementation of each training module and its required parameters.

Regarding passing parameters...

How do we determine the specific values of these parameters, then **effectively** modify and pass them to each module?

1. Starting Training Scripts

- We can directly assign values to various parameters in the training script.
- When the training script is very complex, the various functional modules may become hard to keep track of.
- We usually use a `bash` script to carry the required parameter values and start the execution of the `python` training script.



1. Starting Training Scripts

```bash
#!/bin/bash
python /home/kuangph/CS336-Assignment1/cs336_basics/run_clm.py \
    --d_model 512 \
    --num_heads 8 \
    --d_ff 1344 \
    --vocab_size 32000 \
    --num_layers 8\
    --max_seq_length 256 \
    --seq_length 256 \
    --batch_size 48 \
    --theta 100000 \
    --device cuda \
    --num_epochs 5.5 \
    --lr 1e-4 \
    --lr_min 1e-5 \
    --warmup_ratio 0.05 \
    --warmfix_ratio 0.9 \
    --chunk_size 500000 \
    --vocab_path /home/kuangph/CS336-Assignment1/data/vocab_32000.txt \
    --merges_path /home/kuangph/CS336-Assignment1/data/merges_32000.txt \
    --special_tokens "<|endoftext|>" \
    --corpus_size "2G" \
    --log_interval 20 \
    --save_interval 500 \
    --weight_decay 0.01 \
    --betas 0.9 0.95 \
    --eps 1e-8 \
    --max_norm 1.0
```

- It is equivalent to entering a long command line in the terminal, containing information we need
- For `run_clm.py`, we need to properly parse the command-line arguments.

2. Parameter Decoding

```python
def parse_bash_args():
    parser=argparse.ArgumentParser()

    parser.add_argument("--d_model",type=int,default=512)
    parser.add_argument("--num_heads",type=int,default=8)
    parser.add_argument("--d_ff",type=int,default=1344)
    parser.add_argument("--vocab_size",type=int,default=32000)
    parser.add_argument("--num_layers",type=int,default=6)
    parser.add_argument("--max_seq_length",type=int,default=512)
    parser.add_argument("--seq_length",type=int,default=256)
    parser.add_argument("--batch_size",type=int,default=32)
    parser.add_argument("--theta",type=int,default=100000)
    parser.add_argument("--device",type=str,default="cuda")

    parser.add_argument("--num_epochs",type=float,default=10)
    parser.add_argument("--lr",type=float,default=1e-4)
    parser.add_argument("--lr_min",type=float,default=1e-5)
    parser.add_argument("--warmup_ratio",type=float,default=0.1)
    parser.add_argument("--warmfix_ratio",type=float,default=0.9)
    parser.add_argument("--corpus_size",type=str)
```

```python
    parser.add_argument("--chunk_size",type=int,default=500000)
    parser.add_argument("--vocab_path",type=str,
                        default="data/vocab_32000.txt")
    parser.add_argument("--merges_path",type=str,
                        default="data/merges_32000.txt")
    parser.add_argument("--special_tokens",
                        type=str, nargs="*", default=["<|endoftext|>"])

    parser.add_argument("--log_interval",type=int)
    parser.add_argument("--save_interval",type=int)

    parser.add_argument("--weight_decay",type=float,default=0.01)
    parser.add_argument("--betas",type=float, nargs="*",
                        default=(0.9,0.95))
    parser.add_argument("--eps",type=float,default=1e-8)

    parser.add_argument("--max_norm",type=float,default=1.0)

    args=parser.parse_args()
    return args
```

2. Parameter Decoding

```
[CODE EXPUNGED]
```

```
[CODE EXPUNGED]
```

PART7: Complete Script

Review of Model Training Process

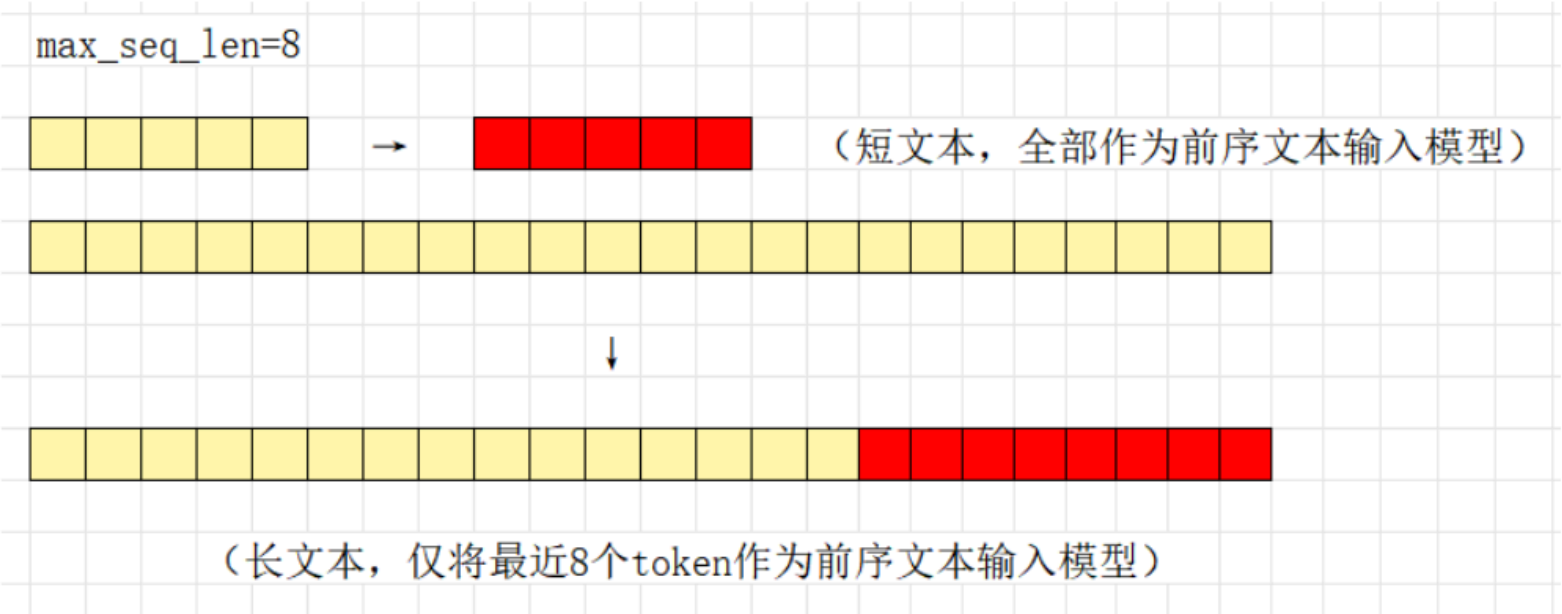
- Receive parameters, set up the model and training modules.
- Determine the total number of iterative optimization steps.
- For each step:
  - Sample data.
  - Forward.
  - Calculate loss.
  - Calculate gradients and clip.
  - Schedule the learning rate.
  - Update parameters.
- It is recommended to output real-time status during the process, such as the current loss function value, current text prediction results, etc.

Code Explanation

PART8: Autoregressive Decoding

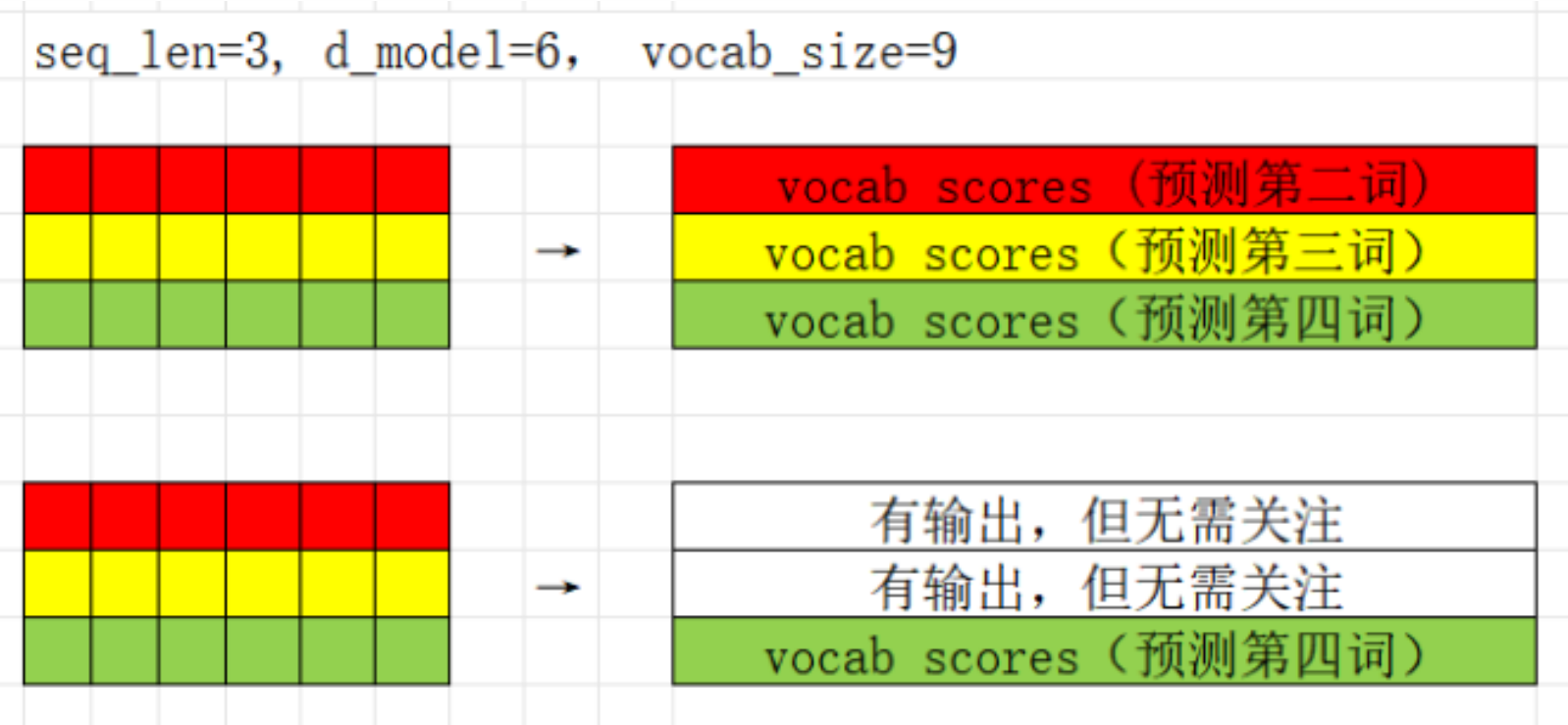
Determining Input Text

During training, fixed-length text is often sampled as input to the model. However, the total length of decoded text may be unpredictable.



Sampling Output Values

- During training, we need to know the output at all positions to calculate the loss value;
- During decoding, we only care about "what is the next word?"!



From Vocab Scores to Token Sampling

After receiving vocab scores:

- Repetition penalty: Reduce scores of recently appeared tokens to avoid repeating the same word.
- Temperature sampling: Scale all scores by the same factor to make softmax differences larger/smaller.
- Softmax: Convert scores into a probability distribution.
- Randomly sample the next token according to the probability distribution.

Autoregressive

The sampled output of this step is part of the input for the next step.

Code Explanation and Practical Demonstration